

Data Card

Dataset Plan for Localized Agronomic Knowledge Retrieval

The dataset plan below is grounded in the Agricultural Extension RAG Retrieval Benchmark, a dataset provided on Kaggle by the TRI AI team for a retrieval competition. That dataset is used here as a reference point, not as a finished solution to the problem described in the accompanying problem statement. Where the existing dataset falls short of what the Nigerian-focused system would actually need, this is stated directly rather than glossed over.

Input vs. Output

What are users inputting? What are the outputs or labels the model will predict or generate?

The input is a natural language query written by a farmer or an extension worker, describing an observed problem such as crop symptoms, soil condition, or pest presence (for example, "why are my maize leaves turning yellow"). The output is not a generated answer but a ranked list of the five most relevant documents drawn from a fixed knowledge base of agricultural extension factsheets, ordered from most to least relevant. This is the retrieval half of a RAG pipeline; a separate generation step would later use these documents to compose a final answer, but that step is outside the scope of this dataset.

Classification Problem

What is the classification task, if any? What categories or classes are you working with, and why?

This is not a classification problem, and framing it as one would misrepresent the task. The actual task is document retrieval and ranking: given a query, the model must retrieve and order the top five documents from a corpus of 695 agricultural extension documents by relevance. Relevance itself is graded rather than a fixed set of classes, scored from 0 to 3 (0 is not relevant, 1 is marginal, 2 is relevant, 3 is a direct answer). The evaluation metric, nDCG@5, reflects this: it rewards both finding relevant documents and ranking the best ones first, which a simple classification framing would not capture.

Labeling

*Where do your labels come from: domain experts, community members, crowd workers, AI-generated sources?
How will you check for consistency and accuracy?*

In the reference dataset, relevance labels come from a graded scoring file covering 4,194 query-document pairs, but the source materials do not disclose who produced these labels, how many annotators were involved, or how agreement between annotators was checked. This is a real gap. For the Nigerian-focused version of this system, labeling would need to come from domain experts, agronomists or extension officers familiar with Nigerian crops and farming conditions, reviewing candidate documents against real farmer queries. At least two independent raters per query, with a documented process for resolving disagreements, would be needed to check consistency and accuracy, since the current dataset offers no such record.

Data Creation

How will you collect your data? Will it come from direct sources, or will you synthesize data in low-resource settings?

The reference corpus is built from two sources: documents authored specifically for the competition, and documents derived from openly licensed materials such as CGIAR and CGSpace publications, rewritten with the help of language models. Neither source involves direct collection from Nigerian farmers or extension workers. For a system meant to serve Nigerian smallholder farmers specifically, this dataset would need to be supplemented with genuine field sources: real extension documents from Nigerian agencies, and real farmer queries gathered through interviews, extension call logs, or community outreach, rather than relying only on synthetic or LLM-rewritten text. The existing corpus targets Sub-Saharan Africa broadly, which is not the same as being representative of Nigeria.

Web Collection Ethics

Do you plan to source data from the web? If yes, which sites or platforms? What tools will you use, AI tools, human workers, or both? Have you addressed consent, copyright, and privacy?

Part of the reference corpus already draws on openly licensed web content, chiefly from CGIAR and CGSpace, under CC-BY or CC0 licenses. Sources marked NoDerivatives or NonCommercial were excluded during filtering, and each document records its source and license. Language models were used to rewrite some of this grounded material. If this dataset were extended with Nigerian sources, the same standard would need to apply: using only openly licensed or explicitly permitted material, keeping attribution intact for every document, and being transparent about which content is LLM-generated rather than expert-authored, so it is not mistaken for verified professional guidance. No personal or private farmer data would be scraped from the web; any farmer queries used would need to be collected with informed consent.

Bias and Representation

Who might be left out of your dataset? How will you ensure the dataset reflects the diversity of the community and includes underrepresented voices?

The reference dataset is English-only, but many Nigerian smallholder farmers are more comfortable communicating in Hausa, Yoruba, Igbo, or Nigerian Pidgin. As it stands, anyone who does not query the system in English is effectively excluded from meaningful use, even though they are the intended beneficiaries. The corpus is also relatively small at 695 documents and broad in geographic scope, covering Sub-Saharan Africa generally rather than being tuned to Nigeria's specific crop mix, agro-ecological zones, or local pest and disease terminology. This means Nigerian-specific concerns could be underrepresented relative to more widely documented regions. Closing this gap would require adding local-language queries and documents, and having the corpus reviewed by Nigerian agronomists rather than assuming regional coverage is sufficient.

Values and Community Alignment

How does your dataset design reflect the values you identified previously? How will your community benefit, and how will you ensure accountability to them?

The dataset design reflects two values from the problem statement: practical accessibility, prioritizing how farmers actually describe problems over exact keyword matching, and rigor, evaluating against expert-labeled relevance judgments rather than assumptions about what a good result looks like. Both align with the goal of closing the gap between where agronomic knowledge exists and where it is actually usable. The community benefits directly when extension workers and farmers spend less time searching for guidance and more time acting on it. Accountability would be maintained by keeping the source and license of every document traceable, and by stating clearly that the system supports rather than replaces professional agronomic advice, since synthetic and LLM-rewritten content can still contain factual errors.

Source note: the dataset referenced throughout this data card is the Agricultural Extension RAG Retrieval Benchmark, provided on Kaggle by the TRI AI team for a retrieval competition. It is used here as a reference point for identifying what a locally relevant version of this dataset would still need.